REVISION NOTES

Week 6

Sun 22 – Fri 27 November 2026

12	10 h 14 min	11 / 12
LECTURES	VIDEO	SUMMARISED

COURSES THIS WEEK

MATH 120	Single-Variable Calculus	2 lectures · 1 h 38 min
AI 200	Neural Networks: Zero to Hero	1 lecture · 3 h 31 min
SPCH 100	Storytelling	4 lectures · 1 h 11 min
MATH 130	Probability	2 lectures · 1 h 33 min
MATH 110	Linear Algebra	2 lectures · 1 h 39 min
AI 210	Practical Deep Learning	1 lecture · 42 min
Total		**12 lectures · 10 h 14 min**

Study days Sun 22 Nov to Fri 27 Nov. Hours are video time; a lecture the plan spreads over several days counts only the days that fall in this week.

WEEK 6 · SUN 22 – FRI 27 NOVEMBER 2026

Contents

LECTURES, IN PLAN ORDER

1	MATH 120	Exam 1 Review		Sun 22
2	AI 200	Deep dive into LLMs like ChatGPT	No summary	Sun 22 – Thu 26
3	SPCH 100	The five dials — rate, volume, melody, tone, pause		Sun 22
4	MATH 130	The Poisson distribution		Mon 23
5	SPCH 100	Melody, volume, and the descending scale		Mon 23 – Tue 24
6	MATH 110	Independence, Basis, and Dimension		Tue 24
7	MATH 120	Linear and Quadratic Approximations		Wed 25
8	SPCH 100	The pause — tie it to a breath		Wed 25
9	MATH 130	Discrete vs. Continuous, the Uniform		Thu 26
10	SPCH 100	Five body-language habits that read as confidence		Thu 26
11	MATH 110	The Four Fundamental Subspaces		Fri 27
12	AI 210	Lesson 1: Getting started		Fri 27

AT THE BACK

Not yet summarised	1 lecture
What this week established	
Answers to the self-checks	

11 of 12 lectures have a summary. The other 1 are listed, with their video links, on “Not yet summarised”; nothing is written for them here.

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 1 OF 12

Exam 1 Review

Sun 22 Nov · 51 min video · Watch on YouTube

THE TAKEAWAY

Exam 1 is Unit I in two columns: five general rules that combine functions, and a memorised list of specific derivatives — plus the ability to read the definition of the derivative backwards.

THE ARGUMENT · 9 STEPS

1. **The five general rules**
   Sum, constant multiple, product, quotient, chain — and implicit differentiation as a sixth technique rather than a formula. Inverse functions and logarithmic differentiation are both implicit differentiation wearing different clothes.

2. **The specific list you must simply know**
   x^r (now for every real r, not just rational), $\sin$, $\cos$, $\tan$, $\sec$, e^x, $\ln x$, $\arctan$, $\arcsin$. Cosecant and cotangent are not on the list because everything reduces to these.

3. **x^r for irrational r, both ways**
   Base e: $x^r = e^{r \ln x}$, chain rule, done. Logarithmic: $\ln u = r \ln x$, so $u'/u = r/x$. Both give rx^{r-1}, and the proof never cared whether r was rational.

4. **The chain rule in one sentence**
   If y moves 10 times as fast as x and x moves 5 times as fast as t, then y moves 50 times as fast as t. Substituting $10(5t + a) + b = 50t + \ldots$ makes it algebra; rates multiply.

5. **You can dodge the quotient rule**
   $(1/v)' = (v^{-1})' = -v^{-2}v'$ by the chain rule — which is also where the minus sign in the quotient rule comes from. $\sec x = (\cos x)^{-1}$ differentiates to $\sin x / \cos^2 x$, conventionally written $\sec x \tan x$.

6. **Pick the rule before you start writing**
   $(x^{10} + 8x)^6$ expanded is a page of exam paper; by the chain rule it is $6(x^{10} + 8x)^5(10x^9 + 8)$ on one line. Do not simplify unless asked — any correct form is a correct answer.

7. **Which derivatives you may be asked to derive from the definition**
   The ones actually done that way in lecture: $1/x$, x^n, $\sin$, $\cos$, a^x, and the product and quotient rules themselves. Nothing requires going all the way back to first principles for the rest.

8. **Read the definition backwards to kill a limit**
   $\lim_{u \to 0} \frac{e^u - 1}{u}$ is not a limit to be fought — it is $\frac{d}{du} e^u$ at $u = 0$, which is 1. Any difference quotient in disguise can be evaluated by recognising whose derivative it is.

9 **Differentiability means the two tangents agree**

There is only one test: compute the left and the right tangent and check they are equal. Sketching y' underneath y is the geometric half — the derivative of $\ln x$ is $1/x$, and the two graphs look nothing like each other.

WORKED EXAMPLE

The promised hardest one: $\frac{d}{dx} e^{x \arctan x}$. Chain rule first, giving $e^{x \arctan x}$ times the derivative of the exponent; then the product rule on $x \arctan x$, giving $\arctan x + \dfrac{x}{1+x^2}$. Four of the listed formulas in one expression, and it fits on a line.

WATCH FOR

Expecting the graph of f' to resemble the graph of f. It does not — it records the slope at each point, so a curve that rises ever more slowly has a derivative that falls towards zero while staying positive.

CHECK YOURSELF · ANSWERS AT THE BACK

1.1 The cleanest way to evaluate $\lim_{u \to 0} \dfrac{e^u - 1}{u}$ is to:

A Apply L'Hopital twice
B Recognise it as the derivative of e^u at $u = 0$, which is 1
C Expand e^u as a product
D Substitute $u = 1/k$ and take $k \to \infty$

1.2 The derivative of $e^{x \arctan x}$ at $x = 1$, to three decimal places, is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 2 OF 12

Deep dive into LLMs like ChatGPT

Sun 22 Nov – Thu 26 Nov · day 1–5 of 5 · 211 min video · Watch on YouTube

NOT YET SUMMARISED

This lecture has no summary yet, so nothing is printed for it. It is listed with its video link on “Not yet summarised” at the back.

SPCH 100 · STORYTELLING · LECTURE 3 OF 12

The five dials — rate, volume, melody, tone, pause

Sun 22 Nov · day 3 of 3 · 43 min video · Watch on YouTube · began in week 5

THE TAKEAWAY

Vinh Giang's five foundations — rate, volume, pitch as melody, tonality, and the pause — coached live on a student, then applied to accents, interviews, imitation and nerves. The voice is behaviour, so it can change.

THE ARGUMENT · 6 STEPS

1. **You are not trapped in your voice**
   Changing your mouth movements and breath changes how you sound. The way you speak is a set of behaviours.

2. **The five dials**
   Vary the rate. Speak at a five, not a three. Use melody, because a song sticks and a page does not. Let your face carry the emotion. Pause.

3. **Live coaching**
   A student answers 3-2-1 style, then again with more volume, gestures and a lower pitch at the end of sentences, then with long pauses and no opening 'so'.

4. **Accent**
   Record twenty improvised minutes, send it to a speech pathologist for pronunciation and to an ESL teacher for grammar, then fix one thing at a time.

5. **Interviews and imitation**
   Spend 5–10 per cent of the time on connection: answer 'tell me about yourself' with a prepared story. Copy great speakers the way a chef copies recipes, then add your own lemon.

6. **Nerves and range**
   Burn off adrenaline with a quick walk or push-ups, use breathing, and focus on the audience. When an unused part of your voice feels fake, tell yourself it is only unfamiliar.

WORKED EXAMPLE

His interview answer: a prepared story (selling MP3 players at 13, and what it taught him about selling the wrong way and the right way), linked to the job, takes about three minutes and builds rapport before the questions start.

WATCH FOR

Calling an unfamiliar voice 'fake'. It is the same instrument — you have just never used those keys, and calling them fake stops you exploring.

CHECK YOURSELF · ANSWERS AT THE BACK

3.1 Why does Giang compare a song with a page of a book?

A Songs are shorter, so the listener can hold the whole thing in their head
B Both hold about as many words, but the melody makes the song memorable
C A song repeats its chorus, so repetition is what makes it stick
D Reading a page aloud is harder than singing, so it sounds flat

3.2 What does he say causes a monotone voice?

A Speaking too fast to let the pitch move
B Too much volume, which flattens the melody
C A monotone body — including a neutral face
D Reading from notes instead of talking

3.3 What is his advice for someone worried about their accent?

A Have a speech pathologist and an ESL teacher review twenty improvised minutes
B Copy one native speaker's accent exactly, sound by sound, until it is yours
C Slow right down and fix every sound at once, so the accent fades in weeks
D Keep quiet in meetings until a coach says your accent has become neutral

MATH 130 · PROBABILITY · LECTURE 4 OF 12

The Poisson distribution

Mon 23 Nov · 43 min video · Watch on YouTube

THE TAKEAWAY

The Poisson is what you get when a LARGE number of unlikely, weakly dependent things could each happen — and the only number you need is λ, the expected count.

THE ARGUMENT · 14 STEPS

1 **First, the mistake this whole course keeps catching**
Confusing a random variable with its distribution — call it sympathetic magic. Adding random variables is not adding PMFs; cubing a random variable is not cubing its PMF. The map is not the territory; the blueprint is not the house.

2 **Why that analogy is the better one**
One blueprint builds many houses. So many different random variables can share a distribution — independent or dependent, it makes no difference to the blueprint. The distribution specifies the probabilities; the random variable is one actual outcome-to-number function.

3 **The PMF**

$P(X = k) = \dfrac{e^{-\lambda}\lambda^k}{k!}$ for $k = 0, 1, 2, \ldots$, with $\lambda > 0$ a rate parameter. Unlike the binomial there is no upper bound on k.

4 **It is valid because of the Taylor series**

$\sum_k \lambda^k/k! = e^\lambda$, so the terms sum to $e^{-\lambda}e^\lambda = 1$. The PMF is literally one term of the series for e^λ, normalised — which is all the distribution is, structurally.

5 **And the mean is λ**

$\sum_k k e^{-\lambda}\lambda^k/k!$: the k cancels one factor of $k!$, pull out a λ, and what is left is the same Taylor series again. Almost nothing needs memorising in this course, but this one does and it is easy.

6 **What it is for**

Counting. Emails in an hour, chips in a cookie, earthquakes in a year in some region. In each case there are very many opportunities and each one is individually unlikely — many people who COULD email you, each unlikely to in this particular hour.

7 **None of those is exactly Poisson**

Most have an obvious upper bound and the Poisson has none, and independence is usually only approximate. It is a first model to check against data, not a claim about the world. That it is nonetheless the most-used discrete distribution in practice is the point.

8 **The Poisson paradigm**

Given events $A_1, \ldots, A_n$ with $P(A_j) = p_j$, n large, all p_j small, and the events independent OR WEAKLY DEPENDENT, the number that occur is approximately Poisson with $\lambda = \sum_j p_j$. That λ is exact by linearity even when the approximation itself is not.

9 **Which is much more than the binomial case**

The binomial needs independent trials with one common p. The paradigm allows different p_j and some dependence — knowing about A_1 and A_2 may shift A_3 slightly. Weak dependence has no clean definition, which is honest: independence comes in degrees.

10 **Binomial converges to Poisson**

Hold $\lambda = np$ fixed, let $n \to \infty$ and $p \to 0$. Write $\binom{n}{k}$ as $n(n-1)\cdots(n-k+1)/k!$; the k factors of n cancel term by term, $(1-\lambda/n)^{-k} \to 1$, and $(1-\lambda/n)^n \to e^{-\lambda}$ by the compound-interest limit. What is left is exactly the Poisson PMF.

11 **Raindrops**

Divide a sheet of paper into millions of tiny squares. Each is very unlikely to be hit in a given minute, and there are enormously many of them. Binomial would need independence and at most one drop per square; Poisson needs neither, and a binomial with n in the trillions is unusable on a computer anyway.

12 **Triple birthday matches**

Among n people, what is the chance three share a birthday? Exactly, this is horrible. There are $\binom{n}{3}$ triplets; each matches with probability $1/365^2$; so $\lambda = \binom{n}{3}/365^2$ by linearity, exactly.

13 **And the approximation finishes it**

The triplet indicators are dependent — I_{123} and I_{124} share two people — but only weakly, since even a head start still needs the fourth person to match. So $P(\text{at least one}) \approx 1 - e^{-\lambda}$, which is a calculator keystroke instead of a page of binomial coefficients.

14 **Why n need not be huge**

As with the ordinary birthday problem, what matters is not n but $\binom{n}{3}$. Twenty people give 1140 triplets. The number of TRIALS is large even when the number of people is not.

WORKED EXAMPLE

When you are counting rare events among many chances, do not fight for the exact answer. Compute λ exactly by linearity, then read the probability of no events off $e^{-\lambda}$ and complement it.

WATCH FOR

Demanding independence before invoking the Poisson. The paradigm is built to tolerate different probabilities and mild dependence; insisting on the binomial's assumptions throws away most of its usefulness.

CHECK YOURSELF · ANSWERS AT THE BACK

4.1 The approximate probability that some three of 100 people share a birthday, to three decimal places, is:

Your answer: _____

4.2 For $X \sim \text{Bin}(1000, 0.002)$, the exact value of $P(X = 3)$, to four decimal places, is:

Your answer: _____

4.3 The Poisson paradigm requires that:

A The events be independent and equally likely
B There be many events, each unlikely, at most weakly dependent
C The number of events be bounded
D Each event have probability exactly λ/n

SPCH 100 · STORYTELLING · LECTURE 5 OF 12

Melody, volume, and the descending scale

Mon 23 Nov – Tue 24 Nov · day 1–2 of 2 · 36 min video · Watch on YouTube

THE TAKEAWAY

Roger Love, interviewed on MarieTV: the voice you have is one you imitated, monotone and timid volume are learned, the falling pitch at every full stop makes people sound sad, and nose breathing saves the voice.

THE ARGUMENT · 5 STEPS

1. **The voice you imitated**
   We copied our parents to connect with them. That outdated voice may now be holding you back, and you can build a new one.

2. **Emotion first**
   He argues that sounds carry the emotion that gets a message remembered, and that great songwriters are storytellers who move you from emotion to emotion in three minutes.

3. **Monotone and volume**
   Most people use one or two notes. Loud on one note sounds angry; loud with melody sounds happy.

4. **The descending scale**
   Dropping in pitch at every comma and full stop, as children are taught, sounds sad. He recommends lifting instead.

5. **Characters and breath**
   You need different voices for work and home. Close your lips and breathe through your nose at the full stops so the voice lasts.

WORKED EXAMPLE

The voicemail test: record a greeting, listen back, and notice what you settle for. That settled-for voice is the one everyone hears. Keep recording until the melody goes up and down.

WATCH FOR

The coaches in this unit disagree. Giang tells a student to end sentences lower for authority; Love says lift the ends for warmth; Treasure warns against making every statement sound like a question. Use all three: lift inside the sentence for warmth, land the key claim low, and never end every line on a question.

CHECK YOURSELF · ANSWERS AT THE BACK

5.1 According to Love, why does loud speech sometimes sound angry?

A Volume always sounds angry, so drop it whenever you can
B It is loud and stays on one note — with melody it sounds happy
C Loud voices push the pitch up, and a high pitch always sounds tense
D It is loud and fast — slowing down makes the same volume calm

5.2 What is the 'descending scale'?

A A singing exercise that walks the voice down from the high notes to low
B Getting quieter and quieter toward the end of every sentence you say
C Dropping pitch at every comma and full stop, which he says sounds sad
D Slowing down at the end of a talk so the last point lands heavily

5.3 What does he say is the main reason speakers lose their voice?

A Breathing in through the mouth
B Speaking too loudly for too long
C Not warming up before speaking
D Talking too fast without pausing

MATH 110 · LINEAR ALGEBRA · LECTURE 6 OF 12

Independence, Basis, and Dimension

Tue 24 Nov · 50 min video · Watch on YouTube

THE TAKEAWAY

A basis is the set of vectors that is neither redundant nor short, and every basis for a given space turns out to have the same size.

THE ARGUMENT · 6 STEPS

1. **More unknowns than equations guarantees a non-zero solution**
   With $n > m$ there are at most m pivots, so at least one free variable survives. Give it the value 1 and back substitution produces a vector in the null space that is not $\vec{0}$. The whole lecture rests on this.

2. **Independence means the only combination giving $\vec{0}$ is the trivial one**
   Put the vectors in the columns of a matrix and the definition becomes a question about $N(A)$: independent when the null space holds only $\vec{0}$, dependent when it holds anything else. Any set containing $\vec{0}$ is dependent immediately.

3. **Spanning means the combinations reach everything**
   The columns of a matrix span its column space — that is what the column space is. Spanning says the set is large enough; independence says it is not too large. Neither implies the other.

4 **A basis is both at once**

Independent and spanning. Hand someone a basis for a subspace and you have told them everything about it. For $\mathbb{R}^n$, n vectors form a basis exactly when the square matrix holding them is invertible.

5 **There are infinitely many bases, all the same size**

Any invertible 3×3 matrix has columns that are a basis for $\mathbb{R}^3$. What every basis for a space shares is its count — and that number is the space's dimension.

6 **Rank is a dimension, and so is the number of free variables**

The pivot columns are a basis for $C(A)$, so $\dim C(A) = r$. The special solutions are a basis for $N(A)$, so $\dim N(A) = n - r$. A matrix has a rank; a space has a dimension — the words do not swap.

WORKED EXAMPLE

A shortcut worth having: once you know the dimension is 2, any two independent vectors from the space are automatically a basis. If they failed to span, a third independent vector would exist — and that would make the dimension bigger than 2.

WATCH FOR

Saying "the dimension of the matrix" or "the rank of the subspace". A matrix has a rank; a space has a dimension. The rank of A is the dimension of $C(A)$.

KEY CONCEPTS

Independence means no wasted direction

A basis is the ruler you measure with

Rank plus nullity is conserved

CHECK YOURSELF · ANSWERS AT THE BACK

6.1 Three vectors in $\mathbb{R}^2$ are:

A Always independent

B Always dependent, because the 2×3 matrix holding them must have a free variable

C Independent if none is zero

D A basis for $\mathbb{R}^2$

6.2 For $A = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{pmatrix}$, the dimension of the null space is:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 7 OF 12

Linear and Quadratic Approximations

Wed 25 Nov · 47 min video · Watch on YouTube

THE TAKEAWAY

Replacing a hard function by its tangent line near one point is the whole of linear approximation, and the second derivative supplies the curvature the line cannot see.

THE ARGUMENT · 8 STEPS

1. **One formula, and it is the tangent line**
   $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$. The right-hand side is exactly the equation of the tangent at x_0, so the claim is only that a curve looks like its tangent near the point of tangency — and nowhere else.

2. **It is the definition of the derivative, read forwards**
   The derivative says $\Delta f / \Delta x \to f'(x_0)$. Turn that round: for small Δx, $\Delta f \approx f'(x_0)\Delta x$. Multiply out and you have the boxed formula. Average rate and instantaneous rate, swapped.

3. **Always expand about 0, and memorise five**
   With $x_0 = 0$ the formula is $f(x) \approx f(0) + f'(0)x$. That gives $\sin x \approx x$, $\cos x \approx 1$, $e^x \approx 1 + x$, $\ln(1 + x) \approx x$ and $(1 + x)^r \approx 1 + rx$ — each one just $f(0)$ and $f'(0)$ read off a two-column table.

4. **Why $\ln(1 + x)$ and not $\ln x$**
   The logarithm has no tangent line at 0 — it dives to $-\infty$. Shifting to $\ln(1 + x)$ puts the expansion point where the function is well behaved. Same for $(1 + x)^r$: x^r at 0 can have infinite slope.

5. **The right-hand side is the EASY side**
   It looks backwards, because the approximation is longer to write. But $\ln(1.1)$ needs a calculator and $1/10$ does not. That trade — an exact ugly thing for an inexact simple one — is the entire point.

6. **Combine the library instead of differentiating**
   For $e^{-3x}/\sqrt{1 + x}$, write it as $e^{-3x}(1 + x)^{-1/2} \approx (1 - 3x)(1 - \frac{1}{2}x)$ and multiply out. Discard the x^2 term — you already threw away infinitely many others getting here — and the answer is $1 - \frac{7}{2}x$.

7. **A real one: GPS time dilation**
   $T' = T(1 - v^2/c^2)^{-1/2} \approx T(1 + \frac{1}{2}v^2/c^2)$. At $v \approx 4$ km/s the correction is about 10^{-10} — a millimetre of position — and the engineers pre-offset the satellites' broadcast frequency by exactly this. The u^2 term is 10^{-20} and was rightly ignored.

8 **Quadratic approximation adds the curvature**

$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \dfrac{f''(x_0)}{2}(x - x_0)^2$. For $\cos x$ the line $y = 1$ cannot even say which side of 1 the curve is on; $1 - \frac{1}{2}x^2$ is the best-fitting parabola and answers that.

WORKED EXAMPLE

$\ln(1.1)$: linear gives $x = 0.1$. Quadratic gives $x - \frac{1}{2}x^2 = 0.1 - 0.005 = 0.095$, and the true value is $0.0953\ldots$ — the correction buys two more digits for one multiplication.

WATCH FOR

Keeping quadratic debris in a linear answer. When $(1 - 3x)(1 - \frac{1}{2}x)$ produces a $\frac{3}{2}x^2$, it is not a bonus term — it is one of infinitely many quadratic contributions you have already dropped, so keeping it makes the answer less honest, not more.

CHECK YOURSELF · ANSWERS AT THE BACK

7.1 Why is $\sin x \approx x$ also the QUADRATIC approximation of $\sin$ near 0?

A Sine is linear near 0
B $f''(0) = -\sin 0 = 0$, so the x^2 term vanishes
C The quadratic rule does not apply to trig functions
D Because $\cos 0 = 1$

7.2 The coefficient of x in the linear approximation of $e^{-3x}/\sqrt{1+x}$ near 0 is:

Your answer: _____

7.3 The quadratic approximation of $\ln(1.1)$, to three decimal places, is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 8 OF 12

The pause — tie it to a breath

Wed 25 Nov · 13 min video · Watch on YouTube

THE TAKEAWAY

Rebecca Martin on the pause: why it works (time to think, time to process, and a sign of status), how to make it natural (tie it to a breath), and the three places it pays off most (before you start, around your main message, and at transitions).

THE ARGUMENT · 5 STEPS

1 **Why pause**

You get time to think, the audience gets time to process, and it shows you think your words are worth waiting for. Rushing comes across as junior.

2 How

Breathe in, speak at the top of the breath, finish the thought as the breath runs out, then breathe in again. That in-breath is the pause.

3 Practice

Read an article aloud and take a full breath at every full stop.

4 Before you start

Three breaths and eye contact with three people. It feels too long; in practice it becomes one and a half breaths, and the room's attention goes up.

5 Message and transitions

Pause around your key line and stay with the audience while you do. Plan pauses between sections, where moving or a sip of water is fine.

WORKED EXAMPLE

She delivers a line ('we are only scared of how powerful and beautiful we actually are') and holds the silence, breathing and checking in silently with the audience, before moving on.

WATCH FOR

A pause in the wrong place — in the middle of a sentence, or after a weak point. A pause makes whatever comes just before it sound important.

CHECK YOURSELF · ANSWERS AT THE BACK

8.1 What does Martin say a natural pause must be tied to?

A A slide change
B A gesture
C A breath
D A sip of water

8.2 What does she suggest before the first word of an important talk?

A Three breaths and eye contact with three people
B A short joke to relax the room before you begin
C Thanking the organisers and the audience warmly
D Reading the agenda aloud so people know the plan

8.3 What does rushing through your content tell the audience, according to her?

A That you are well prepared and know the material inside out
B That you are of junior status and do not want to take up their time
C That the material is easy, so the audience can follow it quickly
D That you are excited about the topic and want to share every bit of it

MATH 130 · PROBABILITY · LECTURE 9 OF 12

Discrete vs. Continuous, the Uniform

Thu 26 Nov · 50 min video · Watch on YouTube

THE TAKEAWAY

In the continuous world the PMF becomes a probability DENSITY, which is not a probability — you integrate it to get one — and every discrete idea has an exact counterpart with a sum replaced by an integral.

THE ARGUMENT · 17 STEPS

1. **The dictionary**
   Discrete: PMF, $\sum$. Continuous: PDF, $\int$. The CDF $F(x) = P(X \le x)$ appears in both columns unchanged, which is the point of defining it — it needs no separate theory.

2. **Why a PMF will not do**
   For a continuous random variable $P(X = x) = 0$ at every single x. Writing down the PMF would give the constant zero function, which tells you nothing, so the weight has to be described some other way.

3. **Pebbles and mud**
   Discrete probability is pebbles with masses adding to 1. Continuous probability is a mass of mud with total mass 1 smeared over the line. Density is mass per unit length — and mass per unit length is not mass.

4. **The definition**
   f is the PDF of X if $P(a \le X \le b) = \int_a^b f(x)\,dx$ for every a and b. Setting $a = b$ gives 0, which agrees with $P(X = x) = 0$ and is why an interval of positive length is needed.

5. **A density can exceed 1**
   Nothing forbids $f(x_0) = 5$, as long as the total area is still 1. If you find that disturbing you are still reading f as a probability.

6. **What it does mean**
   $f(x_0)\,\varepsilon \approx P(x_0 - \varepsilon/2 \le X \le x_0 + \varepsilon/2)$ for tiny ε — over a small enough interval f is nearly constant, and the integral of a constant is the constant times the length. Multiplying by ε is what converts density back to probability.

7. **Validity**
   $f(x) \ge 0$ and $\int_{-\infty}^{\infty} f = 1$. Exactly the PMF's two conditions with the sum replaced.

8. **Both directions between PDF and CDF**
   $F(x) = \int_{-\infty}^{x} f(t)\,dt$ and $f(x) = F'(x)$ — the two halves of the fundamental theorem of calculus, one for each direction. 'Continuous random variable' here means the CDF is differentiable, not merely continuous.

9 **Variance**
$\mathrm{Var}(X) = E\big((X - EX)^2\big)$. Without the square it is always 0 by linearity. Absolute value would work but is not differentiable at the corner; squares bring the Pythagorean geometry with them, and that is worth keeping.

10 **The computing form**
Expand and use linearity: $\mathrm{Var}(X) = E(X^2) - (EX)^2$. Which also settles which order to square in: $E(X^2) \geq (EX)^2$ always, with equality only when X is a constant.

11 **Standard deviation**
$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$, because squaring changed the units — miles became square miles. Variance is nicer to do algebra with, standard deviation is what you interpret.

12 **The uniform**
'Completely random' on (a, b) has to mean probability proportional to LENGTH: equal intervals, equal probability. So the density is constant, and normalising forces $f(x) = 1/(b - a)$ on the interval.

13 **Its CDF and mean**
$F(x) = (x - a)/(b - a)$ on the interval — linear, as accumulating at a constant rate should be — and $E(X) = (a + b)/2$, the midpoint, which would be alarming if it came out any other way.

14 **LOTUS**
To get $E(g(X))$ you do NOT need the distribution of $g(X)$: $E(g(X)) = \int g(x)f(x)\,dx$, or $\sum g(x)P(X = x)$ in the discrete case. The Law of the Unconscious Statistician, named for looking like what you would write while half asleep. It is true, and it is what makes variance computable.

15 **And so the uniform's variance**
For U uniform on $(0, 1)$: $E(U^2) = \int_0^1 u^2\,du = 1/3$ by LOTUS, and $E(U) = 1/2$, so $\mathrm{Var}(U) = 1/3 - 1/4 = 1/12$.

16 **Universality of the uniform**
Let F be any continuous strictly increasing CDF and U uniform on $(0, 1)$. Then $X = F^{-1}(U)$ has CDF F — because $P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$, the last step because for a standard uniform the probability of an interval IS its length.

17 **Which is why it matters in practice**
A computer can produce uniform random numbers and nothing else. Universality says that is enough: any distribution you want is one inverse CDF away, at least in principle.

WORKED EXAMPLE

In the continuous case, write down the PDF, integrate it for probabilities, differentiate the CDF to recover it, and reach for LOTUS the moment you need the expectation of a function rather than of X itself.

WATCH FOR

Reading the PDF's value as a probability. It is a rate; it can be bigger than 1; only its integral over an interval is a probability.

CHECK YOURSELF · ANSWERS AT THE BACK

9.1 A PDF takes the value 2.5 at some point. This means:

A The PDF is invalid, since probabilities cannot exceed 1
B Nothing is wrong — a density is probability per unit length, and only its integral must equal 1
C The random variable is discrete
D The CDF exceeds 1 there

9.2 The variance of a uniform random variable on $(0, 1)$, to four decimal places, is:

Your answer: _____

9.3 Set $X = -\ln(1 - U)$ with U uniform on $(0, 1)$, so that X has CDF $F(x) = 1 - e^{-x}$. Then $P(X \leq 1)$, to four decimal places, is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 10 OF 12

Five body-language habits that read as confidence

Thu 26 Nov · 8 min video · Watch on YouTube

THE TAKEAWAY

Alex Lyon's five habits that come across as confidence even when you are nervous: eye contact (one thought, one look), an occasional smile, a planted ready position, gestures from a home base, and pauses with a nod.

THE ARGUMENT · 5 STEPS

1. **Eye contact**
   About 90 per cent of the time, three to five seconds per person, everyone in the room several times. Glance at notes and come straight back.

2. **Smile**
   A stone face looks like a deer in the headlights. Smiling relaxes you and warms up your eye contact.

3. **Posture**
   Feet shoulder-width apart, weight forward, knees soft, no swaying. Walk with a purpose, then stop and plant again.

4. **Hands**
   Home base: hands loosely together at the waist, or relaxed at your sides. A small gesture about once a sentence.

5 **Pause and nod**

One second after a sentence, two after a key idea, and a small nod. Audiences tend to nod back.

WORKED EXAMPLE

Practise 'one thought, one look' out loud: say half a sentence to one person, move your eyes as the next half starts, and repeat until it is automatic.

WATCH FOR

The advice to look over people's heads. It may calm you, but nobody feels connected, so it does not look confident.

CHECK YOURSELF · ANSWERS AT THE BACK

10.1 How long does Lyon suggest looking at each person?

A Under a second each, so nobody feels singled out
B About three to five seconds — one thought, one look
C As long as they keep looking back at you, then move
D About ten seconds each, so they feel fully seen

10.2 What is the 'ready position'?

A Feet together, hands clasped in front, weight resting back on the heels
B One foot slightly forward, leaning on the lectern with both hands
C Hands in pockets, weight on the heels, shifting gently side to side
D Feet shoulder-width apart, weight on the balls of the feet, knees soft

10.3 What does he recommend adding to the longer pauses?

A A small nod
B A sip of water
C A glance at the slides
D A smile at the floor

MATH 110 · LINEAR ALGEBRA · LECTURE 11 OF 12

The Four Fundamental Subspaces

Fri 27 Nov · 49 min video · Watch on YouTube

THE TAKEAWAY

A matrix carries four subspaces, and their dimensions are fixed by the rank alone: r, $n - r$, r and $m - r$.

THE ARGUMENT · 6 STEPS

1 **Two spaces in $\mathbb{R}^n$, two in $\mathbb{R}^m$**

The null space and the row space hold vectors with n components; the column space and the left null space hold vectors with m components. The row space is written $C(A^T)$ so everything stays a column.

---

2 **The row space has the same dimension as the column space**

Both are r. That is the lecture's one surprising claim, and it settles questions the columns alone hide: a 3×3 matrix with two equal rows has rank 2, so its three columns cannot be independent, however unrelated they look.

---

3 **Row reduction destroys the column space and preserves the row space**

Every row operation takes combinations of rows, so nothing leaves the row space and the steps reverse. The columns, meanwhile, are genuinely different vectors — $C(R) \neq C(A)$.

---

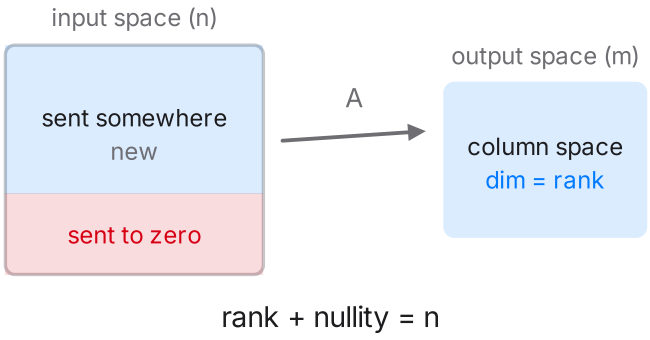
4 **So R hands you the best basis for the row space**

The first r rows of R — of R, not of A. The pivot columns of A — of A, not of R — are the basis for the column space. The two spaces read off opposite matrices, and swapping them is the standard mistake.

---

5 **The special solutions are a basis for the null space**

One per free variable, so $\dim N(A) = n - r$. With $\dim C(A^T) = r$, the two subspaces of $\mathbb{R}^n$ account for all n — which is just the pivot-and-free split of the variables, stated as dimensions.



---

6 **The left null space needs E, not R**

$\vec{y}$ with $A^T\vec{y} = \vec{0}$ is $\vec{y}^T A = \vec{0}^T$ transposed: a combination of ROWS giving the zero row. Run Gauss-Jordan on $[A; |; I]$; the E that appears satisfies $EA = R$, and its last $m - r$ rows are that basis.

WORKED EXAMPLE

Vector spaces need not hold arrows. The 3×3 matrices form one — they add and scale, and nothing asks them to multiply. Upper triangular is a subspace, symmetric is another, and their intersection is the diagonal matrices, of dimension 3.

WATCH FOR

Taking the basis of the row space from A and the basis of the column space from R. It is the other way round: R's rows span the same row space, but R's columns do not span $C(A)$.

KEY CONCEPTS

Rank plus nullity is conserved

A basis is the ruler you measure with

CHECK YOURSELF · ANSWERS AT THE BACK

11.1 Row reduction preserves the row space but not the column space because:

A It changes the rank

B Each step replaces a row by a combination of rows, and the steps reverse

C The columns are never touched

D R has fewer rows than A

11.2 For $A = \begin{pmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{pmatrix}$, the dimension of the left null space is:

Your answer: _____

AI 210 · PRACTICAL DEEP LEARNING · LECTURE 12 OF 12

Lesson 1: Getting started

Fri 27 Nov · day 1 of 2 · 83 min video · Watch on YouTube · finishes in week 7

THE TAKEAWAY

Deep learning is taught here top-down: train a working image classifier in minutes first, then name what happened — Samuel's loop of weights, an automatic measure of performance and an automatic way to improve it — and learn the discipline that keeps it honest: a validation set, and a test set hidden even from yourself.

THE ARGUMENT · 9 STEPS

1 **From joke to two minutes**

In 2015 recognising a bird in a photo was an XKCD punchline. The lesson's notebook searches DuckDuckGo for bird and forest photos, downloads about 200 of each per search, removes broken files, fine-tunes a pretrained resnet18 for three epochs, and then asks it whether the test photo is a bird, and with what probability.

2 **A DataBlock says what the data is**

blocks=(ImageBlock, CategoryBlock): images in, categories out. get_items finds the files; RandomSplitter(valid_pct=0.2, seed=42) holds out 20% for validation; get_y=parent_label takes each label from the folder an image sits in; Resize(192, method='squish') makes every image the same size.

3 **You don't need what you think you need**

High-school maths, sometimes fewer than 50 labelled items, and free GPUs are enough. The book teaches the whole game first — Perkins's baseball, Lockhart's lament about how maths is taught — and theory once you have the context and the motivation.

4 **Neural nets: a long history**

McCulloch and Pitts's artificial neuron (1943); Rosenblatt's Mark I Perceptron; Minsky and Papert showing one layer cannot learn XOR — and that more layers fix it, which was ignored; the 1986 PDP volumes; and only recently more layers, better hardware and more data.

5 **Samuel's definition of machine learning**

Don't write the steps; show examples. A model has weights; an automatic test measures how well a weight assignment performs; an automatic mechanism changes the weights to do better. Today: architecture, parameters, predictions, labels, loss — and a trained model is just a program.

6 **What models can't do**

No data, no model; no labels, no training; a model only predicts, it does not recommend actions; and a model in the world can create a feedback loop, as predictive policing does when its predicted arrests become its next training data.

7 **The cat recogniser, line by line**

is_cat reads the label from the file name; valid_pct=0.2 and seed=42 fix the validation set; resnet34 is the architecture and error_rate the metric — for people, unlike the loss, which is for SGD. A pretrained model gets a new head, and fine_tune fits the head, then the whole model.

8 **Not just photos**

Early layers learn edges and gradients, later ones wheels, text and petals (Zeiler and Fergus). Sound as spectrograms, time series as images, mouse movements, malware bytes — if a person can see the categories, an image model can learn them. The same few lines also do segmentation, sentiment, tabular data and recommendations.

9 **Validation and test sets**

Choosing hyperparameters by looking at validation results overfits the validation set, so keep a test set hidden even from yourself. And pick validation data like the future: the latest dates for time series, unseen drivers or boats when production will see new ones.

WORKED EXAMPLE

Before training anything, write down how the data will be split. Ask what new data will look like in use — later dates, new people, new places — and make the validation set look like that, not like a random sample of what you already have. Then keep a test set that you look at once, at the end, with a metric you chose for what actually matters.

WATCH FOR

Judging a model by its accuracy on the data it was trained on, or tuning hyperparameters against the validation set until it stops being unseen. A big enough model memorises its training set; only data it never influenced says how it will do.

CHECK YOURSELF · ANSWERS AT THE BACK

12.1 In the bird notebook's DataBlock, where does each image get its label?

A From the name of the folder it was saved in, via get_y=parent_label
B From a pretrained model that tags each download as bird or forest first
C From the search result's caption, saved with each image
D From the file name, whose first letter is upper case for each bird photo

12.2 In Arthur Samuel's 1962 description, what has to be automatic for a program to 'learn' from its experience?

A Writing out each step of the task, which the computer then repeats faster
B Testing how well the current weights perform, and changing them to do better
C Storing every example it has seen, so that it can look the answer up later
D Choosing the architecture, after which the weights can be set by hand

12.3 Training minimises a loss, yet fastai reports error_rate after every epoch. Why have both?

A error_rate is measured on the training set, and the loss on the validation set
B They are the same number, which fastai simply prints under two names
C The loss has to suit SGD to update weights; the metric is for people to read
D The loss is only used with pretrained models; the metric is used for the rest

WEEK 6 · SUN 22 – FRI 27 NOVEMBER 2026

Not yet summarised

1 lecture this week has no summary yet. Watch or rewatch them from the links below.

LECTURES

2 Deep dive into LLMs like ChatGPT

AI 200 · Sun 22 Nov – Thu 26 Nov · day 1–5 of 5 · 211 min video

https://www.youtube.com/watch?v=7xTGNNLPyMI

WEEK 6 · SUN 22 – FRI 27 NOVEMBER 2026

What this week established

Each lecture’s takeaway, by course.

MATH 120 · Single-Variable Calculus

1 **Exam 1 Review.** Exam 1 is Unit I in two columns: five general rules that combine functions, and a memorised list of specific derivatives — plus the ability to read the definition of the derivative backwards.

7 **Linear and Quadratic Approximations.** Replacing a hard function by its tangent line near one point is the whole of linear approximation, and the second derivative supplies the curvature the line cannot see.

AI 200 · Neural Networks: Zero to Hero

2 **Deep dive into LLMs like ChatGPT.** Not yet summarised.

SPCH 100 · Storytelling

3 **The five dials — rate, volume, melody, tone, pause.** Vinh Giang's five foundations — rate, volume, pitch as melody, tonality, and the pause — coached live on a student, then applied to accents, interviews, imitation and nerves. The voice is behaviour, so it can change.

5 **Melody, volume, and the descending scale.** Roger Love, interviewed on MarieTV: the voice you have is one you imitated, monotone and timid volume are learned, the falling pitch at every full stop makes people sound sad, and nose breathing saves the voice.

8 **The pause — tie it to a breath.** Rebecca Martin on the pause: why it works (time to think, time to process, and a sign of status), how to make it natural (tie it to a breath), and the three places it pays off most (before you start, around your main message, and at transitions).

10 **Five body-language habits that read as confidence.** Alex Lyon's five habits that come across as confidence even when you are nervous: eye contact (one thought, one look), an occasional smile, a planted ready position, gestures from a home base, and pauses with a nod.

MATH 130 · Probability

4 **The Poisson distribution.** The Poisson is what you get when a LARGE number of unlikely, weakly dependent things could each happen — and the only number you need is λ, the expected count.

9 **Discrete vs. Continuous, the Uniform.** In the continuous world the PMF becomes a probability DENSITY, which is not a probability — you integrate it to get one — and every discrete idea has an exact counterpart with a sum replaced by an integral.

MATH 110 · Linear Algebra

6 **Independence, Basis, and Dimension.** A basis is the set of vectors that is neither redundant nor short, and every basis for a given space turns out to have the same size.

11 **The Four Fundamental Subspaces.** A matrix carries four subspaces, and their dimensions are fixed by the rank alone: r, $n-r$, r and $m-r$.

AI 210 · Practical Deep Learning

12 **Lesson 1: Getting started.** Deep learning is taught here top-down: train a working image classifier in minutes first, then name what happened — Samuel's loop of weights, an automatic measure of performance and an automatic way to improve it — and learn the discipline that keeps it honest: a validation set, and a test set hidden even from yourself.

WEEK 6 · SUN 22 – FRI 27 NOVEMBER 2026

Answers

The answers and explanations are the summaries' own.

1 Exam 1 Review

1.1 B — Recognise it as the derivative of e^u at $u = 0$, which is 1

It is the difference quotient for e^u at 0 written out in full. Reading the definition of the derivative in reverse turns a limit problem into a lookup.

1.2 2.819

$e^{x \arctan x}\left(\arctan x + \frac{x}{1+x^2}\right)$ at $x = 1$ is $e^{\pi/4}(\pi/4 + 1/2) = 2.193 \times 1.285 = 2.819$.

3 The five dials — rate, volume, melody, tone, pause

3.1 B — Both hold about as many words, but the melody makes the song memorable

So speaking with more melody makes what you say stick.

3.2 C — A monotone body — including a neutral face

Your face controls the emotion under your words.

3.3 A — Have a speech pathologist and an ESL teacher review twenty improvised minutes

Then work on one thing at a time, so you do not freeze up.

4 The Poisson distribution

4.1 0.703

$\lambda = \binom{100}{3}/365^2 = 161{,}700/133{,}225 = 1.2137$, so $1 - e^{-\lambda} = 0.703$. A triple match among a hundred people is more likely than not.

4.2 0.1806

The Poisson approximation with $\lambda = np = 2$ gives $e^{-2}2^3/3! = 0.1804$ — wrong in the fourth decimal place, from a distribution far easier to compute with.

4.3 B — There be many events, each unlikely, at most weakly dependent

Different p_j are allowed and mild dependence is allowed. That tolerance is why the Poisson fits real counting data that no binomial describes.

5 Melody, volume, and the descending scale

5.1 B — It is loud and stays on one note — with melody it sounds happy

Volume mixed with melody is heard as energy, not anger.

5.2 C — Dropping pitch at every comma and full stop, which he says sounds sad

He traces it back to how children are taught to read aloud.

5.3 A — Breathing in through the mouth

The nose moistens the air; breathe through it at commas and full stops.

6 Independence, Basis, and Dimension

6.1 B — Always dependent, because the 2×3 matrix holding them must have a free variable

Two rows allow at most two pivots, so one of the three columns is free. Setting it to 1 produces a non-zero combination of the three that gives $\vec{0}$.

6.2 2

Column three is column one plus column two and column four repeats column one, so the rank is 2. Then $n - r = 4 - 2 = 2$, which is also the number of special solutions.

7 Linear and Quadratic Approximations

7.1 B — $f''(0) = -\sin 0 = 0$, so the x^2 term vanishes

The second derivative of $\sin$ is $-\sin$, which is 0 at 0. The first correction to $\sin x \approx x$ is cubic, which is why the approximation is unusually good.

7.2 −3.5

It is $f'(0)$. Multiplying the two library approximations gives $-3 - \frac{1}{2} = -\frac{7}{2}$, without ever running the quotient rule on that expression.

7.3 0.095

$\ln(1+x) \approx x - \frac{1}{2}x^2$ at $x = 0.1$ gives $0.1 - 0.005 = 0.095$, which agrees with the true 0.09531 to three places.

8 The pause — tie it to a breath

8.1 C — A breath

When the thought runs out with the breath, the in-breath is the pause.

8.2 A — Three breaths and eye contact with three people

It settles your nerves and signals that something important is about to happen.

8.3 B — That you are of junior status and do not want to take up their time

Pausing shows you are secure in your role and in the value you bring.

9 Discrete vs. Continuous, the Uniform

9.1 B — Nothing is wrong — a density is probability per unit length, and only its integral must equal 1

A density of 2.5 over an interval of length 0.01 contributes about 0.025 of probability. Height is not area.

9.2 0.0833

$E(U^2) - (EU)^2 = 1/3 - 1/4 = 1/12 = 0.0833$, with $E(U^2)$ obtained from LOTUS without ever finding the distribution of U^2.

9.3 0.6321

$1 - e^{-1} = 0.6321$. Universality of the uniform: inverting that CDF and feeding it a uniform draw manufactures an exponential random variable out of nothing but a random number generator.

10 Five body-language habits that read as confidence

10.1 B — About three to five seconds — one thought, one look

Long enough to finish a thought and for them to feel a connection.

10.2 D — Feet shoulder-width apart, weight on the balls of the feet, knees soft

Plant from the waist down; be animated from the waist up.

10.3 A — A small nod

It looks confident and audiences tend to nod along.

11 The Four Fundamental Subspaces

11.1 B — Each step replaces a row by a combination of rows, and the steps reverse

Every new row is a combination of the old ones and the operations undo, so the spans match exactly. The columns of R are different vectors from the columns of A, so their span differs too.

11.2 1

$m - r = 3 - 2 = 1$. Row three equals row one, so a single combination — row three minus row one — gives the zero row.

12 Lesson 1: Getting started

12.1 A — From the name of the folder it was saved in, via get_y=parent_label

The downloads were saved into bird_or_not/bird and bird_or_not/forest, so the parent folder's name is the label. Reading a label off the first letter of the file name is the chapter 1 pets trick (is_cat), where the dataset's creators encoded cats that way.

12.2 B — Testing how well the current weights perform, and changing them to do better

Samuel's idea: a weight assignment, an automatic means of testing its actual performance, and a mechanism for altering it to maximise that performance. When adjusting the weights is automatic too, the machine 'learns'. Spelling out every step is exactly what machine learning avoids.

12.3 C — The loss has to suit SGD to update weights; the metric is for people to read

The loss is the measure of performance the training system uses to change the weights, so it must be easy for SGD to work with. A metric is defined for human consumption — easy to understand and close to what you care about — and fastai reports it on the validation set.